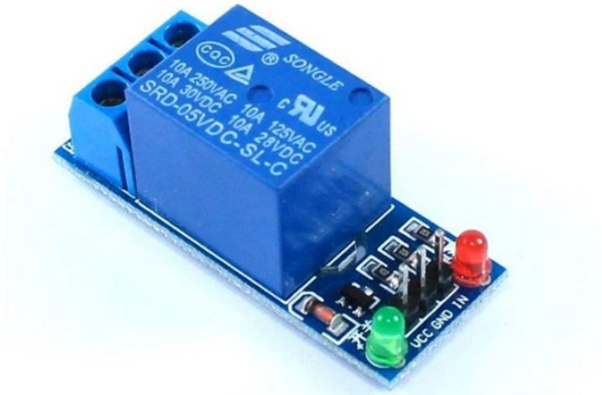
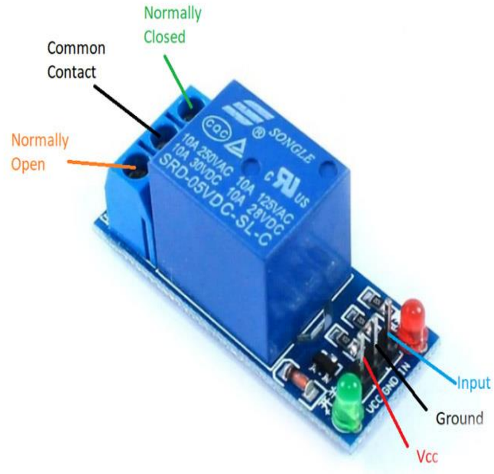




5V Single-Channel Relay Module
Single-Channel Relay Module

Relay is an electromechanical device that uses an electric current to open or close the contacts of a switch. The single-channel relay module is much more than just a plain relay, it comprises of components that make switching and connection easier and act as indicators to show if the module is powered and if the relay is active or not.

Single-Channel Relay Module Pin Description

Pin Number	Pin Name	Description
1	Relay Trigger	Input to activate the relay
2	Ground	0V reference
3	VCC	Supply input for powering the relay coil
4	Normally Open	Normally open terminal of the relay
5	Common	Common terminal of the relay
6	Normally Closed	Normally closed contact of the relay

Single-Channel Relay Module Specifications

- Supply voltage – 3.75V to 6V
- Quiescent current: 2mA
- Current when the relay is active: ~70mA

- Relay maximum contact voltage – 250VAC or 30VDC
- Relay maximum current – 10A

Alternate Relay Modules

Dual-channel relay module, four-channel relay module, 8-channel relay module.

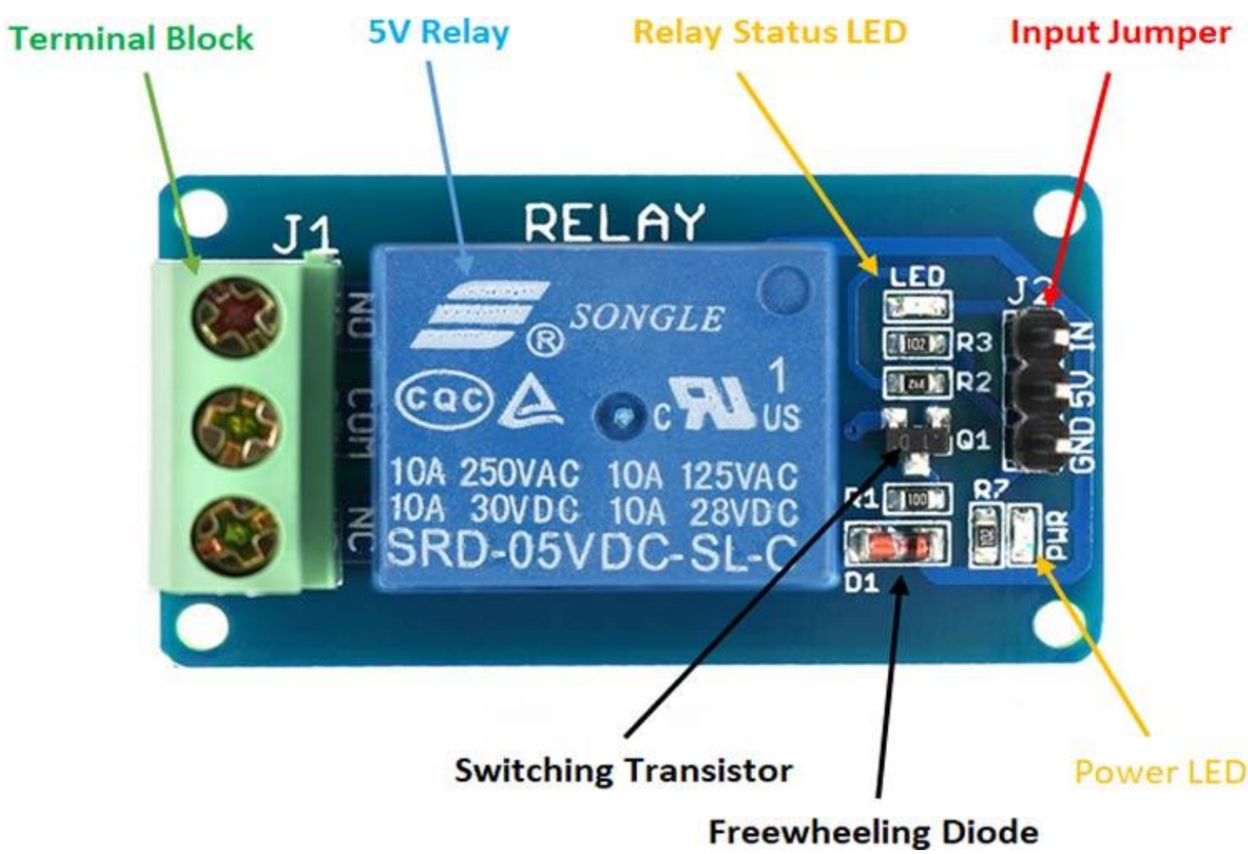
Alternate Switching Modules

Solid State Relay Module, TRIAC, SCR

Components Present on a 5V Single Channel Relay Module

The following are the major components present on a relay module; we will get into the details later in this article.

5V Relay, Transistor, Diode, LEDs, Resistors, Male Header pins, 3-pin screw-type terminal connector, etc.



The single-channel relay module is much more than just a plain relay, it contains components that make switching and connection easier and act as indicators to show if the module is powered and if the relay is active.

First is the screw terminal block. This is the part of the module that is in contact with mains so a reliable connection is needed. Adding screw terminals makes it easier to connect thick mains cables, which might be difficult to solder directly. The three connections on the terminal block are connected to the normally open, normally closed, and common terminals of the relay.

The second is the relay itself, which, in this case, is a blue plastic case. Lots of information can be gleaned from the markings on the relay itself. The part number of the relay on the bottom says "05VDC", which means that the relay coil is activated at 5V minimum – any voltage lower than this will not be able to reliably close the contacts of the relay. There are also voltage and current markings, which represent the maximum voltage and current, the relay can switch. For example, the top left marking says "10A 250VAC", which means the relay can switch a maximum load of 10A when connected to a 250V mains circuit. The bottom left rating says "10A 30VDC", meaning the relay can switch a maximum current of 10A DC before the contacts get damaged.

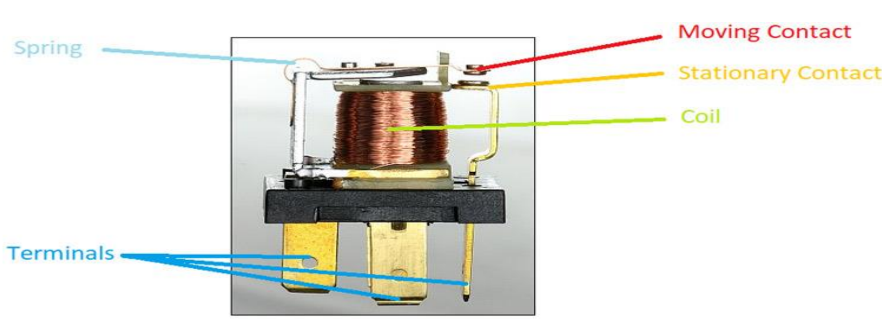
The 'relay status LED' turns on whenever the relay is active and provides an indication of current flowing through the relay coil.

The input jumper is used to supply power to the relay coil and LEDs. The jumper also has the input pin, which when pulled high activates the relay.

The switching transistor takes an input that cannot supply enough current to directly drive the relay coil and amplifies it using the supply voltage to drive the relay coil. This way, the input can be driven from a microcontroller or sensor output. The freewheeling diode prevents voltage spikes when the relay is switched off.

The power LED is connected to V_{cc} and turns on whenever the module is powered.

How Does A Relay Work?

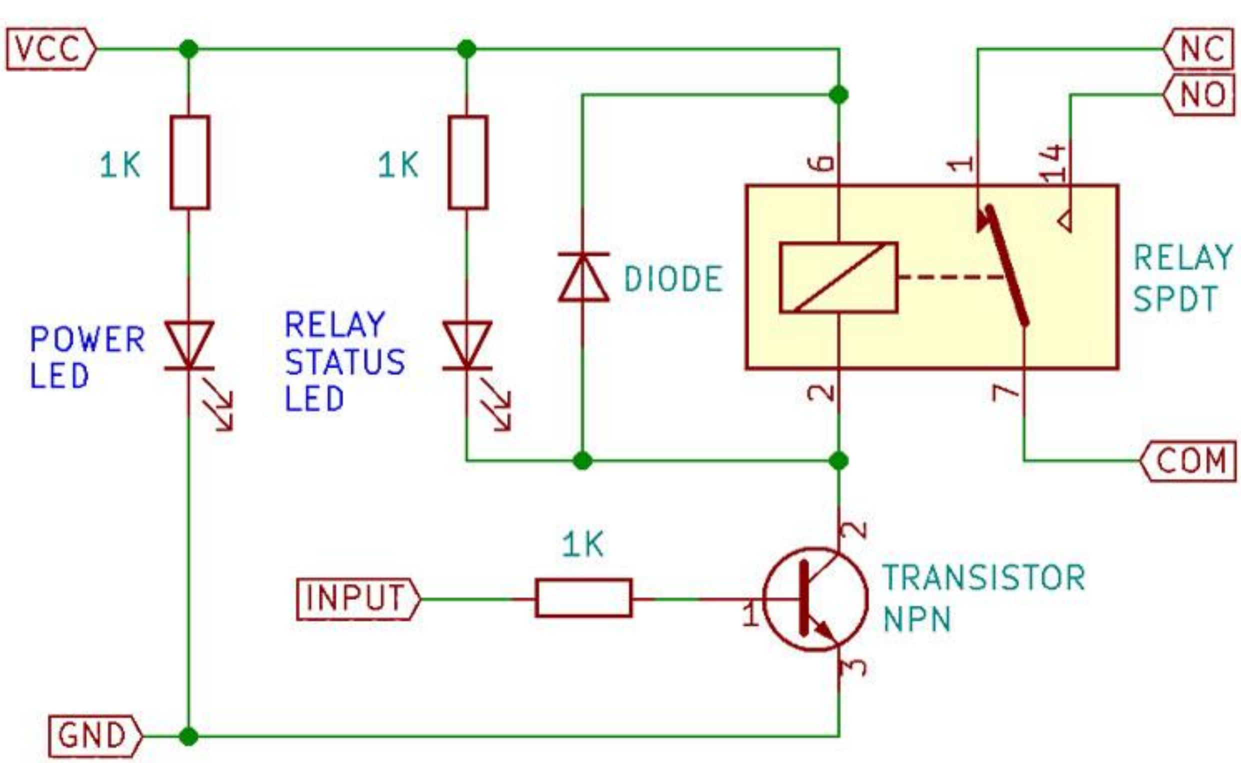


The relay uses an electric current to open or close the contacts of a switch. This is usually done using the help of a coil that attracts the contacts of a switch and pulls them together when activated, and a spring pushes them apart when the coil is not energized.

There are two advantages of this system – First, the current required to activate the relay is much smaller than the current that relay contacts are capable of switching, and second, the coil and the contacts are galvanically isolated, meaning there is no electrical connection between them. This means that the relay can be used to switch mains current through an isolated low voltage digital system like a microcontroller.

Internal Circuit Diagram for Single Channel Relay Module

The circuit on the PCB is quite simple.



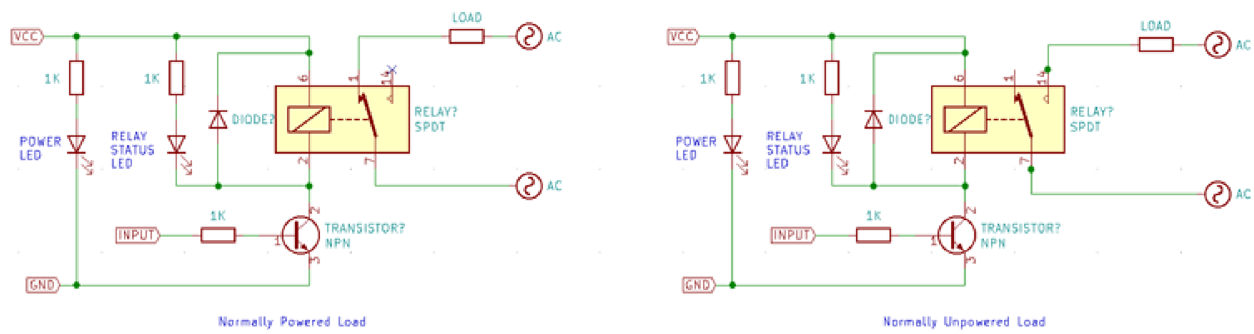
Relay Module Basic Schematic

The extra components apart from the relay are there since it would not be possible to drive the relay directly from the pins of a microcontroller. digital logic or a sensor. This is because although the coil consumes much less current than the currents it can switch, it still needs relatively significant current – low power relays consume around 50mA while higher power relays consume around 500mA. The coil is also an inductive load, so when the coil is switched off, a large flyback voltage is developed which can damage the device turning it on and off. For this reason, a flyback diode is added anti-parallel to the relay coil to clamp the flyback voltage.

LEDs can be added to this basic circuit to act as indicators, and sometimes even optical isolation is added to the input to further improve the isolation.

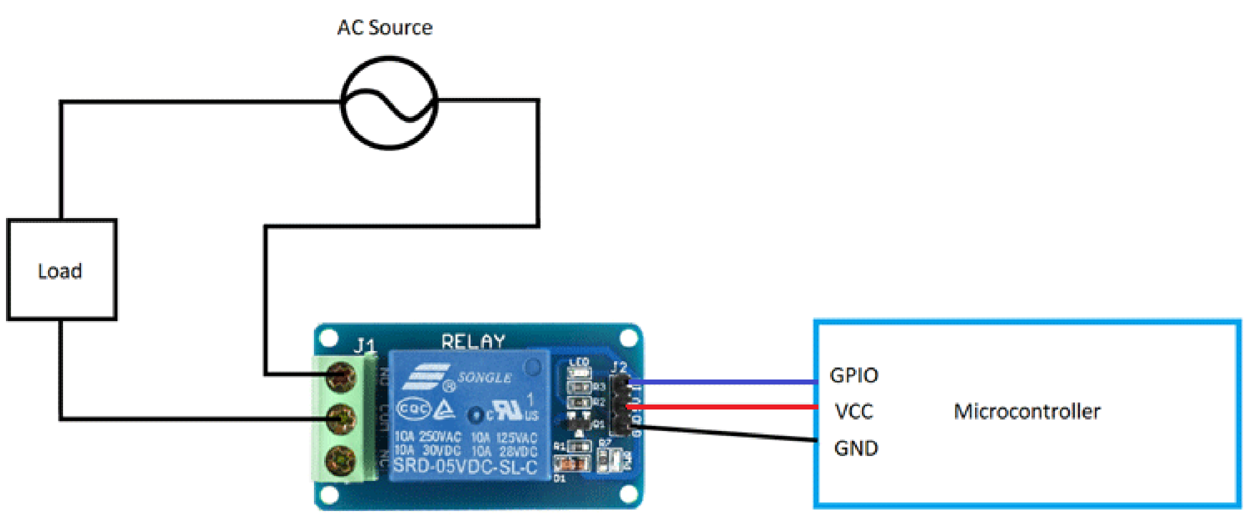
How to use Single-Channel Relay Module

Relay modules like this one are commonly used to drive mains loads from a microcontroller like the Arduino or a sensor. In cases like this, the common circuit diagram would be as follows.



For simple on/off applications, the relay can be connected as shown above. One terminal of mains is connected to common, and the other is connected to NO or NC depending on whether the load should be connected/disconnected when the relay is active.

Check out the image below to see how the relay module is connected to a microcontroller and mains source and load.



The mains wiring is screwed to the terminal block, and the microcontroller can be connected using jumper cables.

Single-Channel Relay Module Basic Trouble Shooting

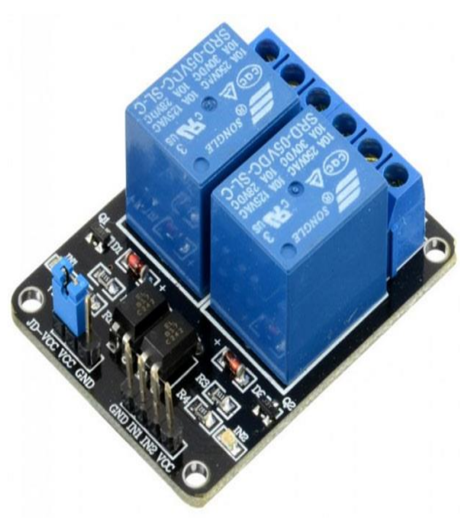
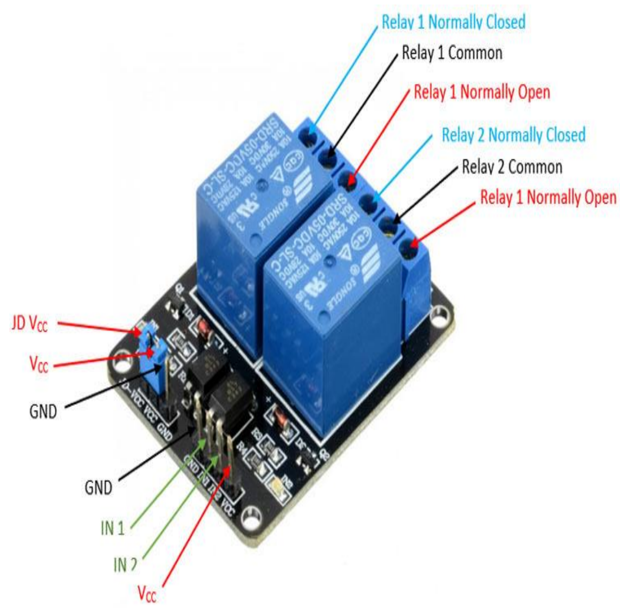
If the relay does not switch on, i.e. no audible clicking sound is heard:

- The contacts might be stuck - Check by physically shaking the relay, if a light clicking sound is not heard, then tap the relay hard, in most cases, this should 'unstick' both the contacts.
- If the contacts do click when the relay is shaken, then the transistor or the flyback diode might be damaged and must be replaced.

Single-Channel Relay Module Applications

- Mains switching
- High current switching
- Isolated power delivery
- Home automation

5V Dual-Channel Relay Module

5V Dual-Channel Relay Module
Dual-Channel Relay Module

The dual-channel relay module is more or less the same as a single-channel relay module, but with some extra features like optical isolation. The dual-channel relay module can be used to switch mains powered loads from the pins of a microcontroller.

Dual-Channel Relay Module Pinout

Pin Number	Pin Name	Description
1	JD-V_{CC}	Input for isolated power supply for relay coils
2	V_{CC}	Input for directly powering the relay coils
3	GND	Input ground reference
4	GND	Input ground reference
5	IN1	Input to activate the first relay
6	IN2	Input to activate the second relay
7	V_{CC}	V_{CC} to power the optocouplers, coil drivers, and associated circuitry

Dual-Channel Relay Module Specifications

- Supply voltage – 3.75V to 6V
- Trigger current – 5mA
- Current when relay is active - ~70mA (single), ~140mA (both)
- Relay maximum contact voltage – 250VAC, 30VDC
- Relay maximum current – 10A

Alternate Relay Modules

Single-channel relay module, four-channel relay module, eight-channel relay module.

Alternate Modules for AC switching

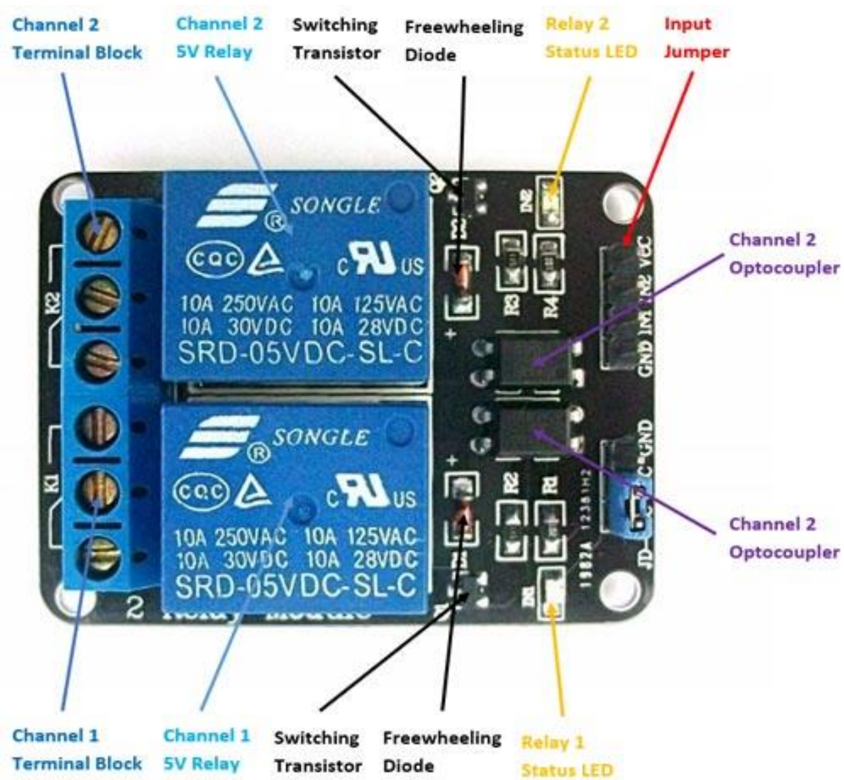
Solid State Relay module, SCRs, TRIACs

Components present on a 5V Dual Channel Relay Module

The following are the major components that would find on a Dual channel Relay module, we will discuss them in detail further in this article.

5V Relay, Optocoupler, Diodes, Transistors, Resistors, LEDs, Male Headers, 3-pin Terminal connectors, etc.

Understanding 5V Dual-Channel Relay Module



The dual-channel relay module contains switching relays and the associated drive circuitry to make it easy to integrate relays into a project powered by a microcontroller. On the left are two terminal blocks, which are used to connect mains wires to the module without soldering.

Next, come to the two relays. As marked on the body of the relay, the relay coil is rated for 5VDC, and the contacts are rated for 10A at 250VAC or 30VDC, or 125VAC or 28VDC.

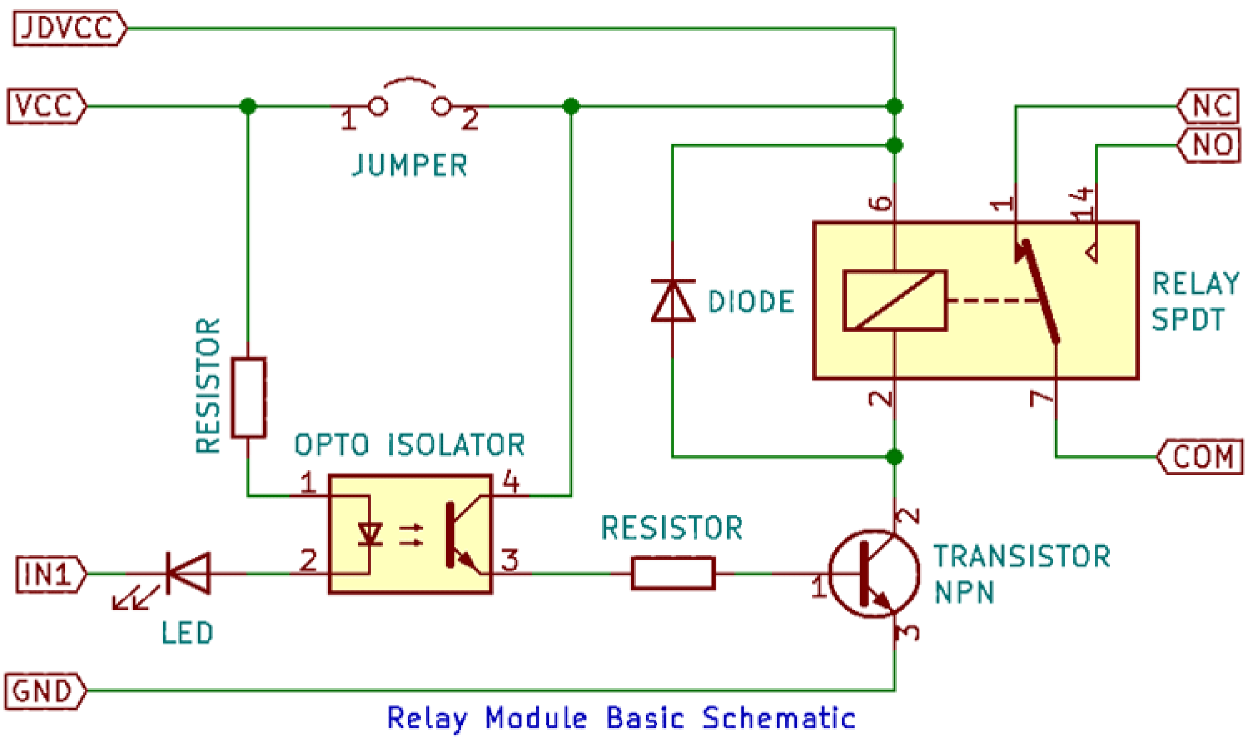
The switching transistors amplify the signal from the inputs enough to drive the relay. The freewheeling diodes prevent voltage spikes across the switching transistors. The status LEDs turn on when the relay is active and indicate switching.

The optocouplers are used to provide additional isolation between the input and the relays. The isolation can be selected using the V_{cc}/JDV_{cc} jumper.

The input jumper has two input and two power pins and can be easily used to connect to jumper wires and other microcontrollers and sensors.

Internal Circuit Diagram for Dual-Channel Relay Module

The circuit on the board is as follows:



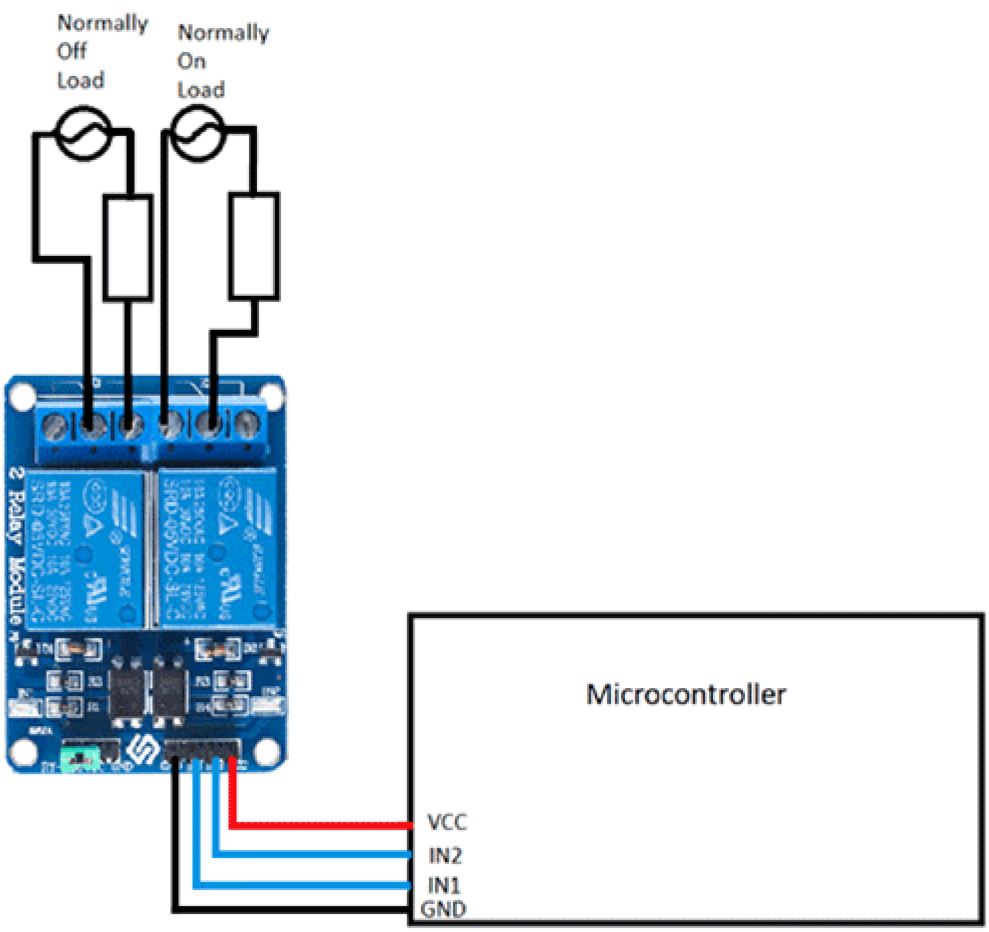
This same circuit is replicated twice on the module. There are several differences when compared to the single-channel relay module. The first is that the optocoupler and relay can have a separate power supply from the input. This is selected using the JD-V_{cc} jumper – if the jumper is shorted, then the relay coil and optocoupler output are connected to the signal V_{cc}. If the jumper is left open, a separate power supply can be provided to the relay coil through the JD-V_{cc} pin.

The second is that the input is active low, meaning that the relay coil is activated only when the input is low. This is because the optocoupler input and the indicator LED are chained to V_{cc}. When the input is high, the voltage difference across the chain is 0V and no current can flow, but when the input is low, supply voltage appears across the chain and the optocoupler output conducts, supplying current to the base of the drive transistor, turning the relay on.

How to use Dual-Channel Relay Module

The dual-channel relay module can be used to switch mains powered loads from the pins of a microcontroller. Since there are two channels on the same board, two separate loads can be powered. This is useful for home automation.

The loads can be connected as follows:



In this diagram, the relay on the left (channel 1) is connected in a regular fashion with the load and source going between common and normally open, so when the relay is activated, the load is powered. On the right channel (channel 2), the load and source are connected between common and normally closed, which means that the load is powered by default till the relay is activated.

If the load is connected to the normally closed contact, then the input polarity 'reversal' can be fixed – the load is powered on when the input is high, and the load is powered off when the input is low. This method, however, wastes some current since the relay coil draws current when the load is switched off.

Dual-Channel Relay Module Basic Troubleshooting

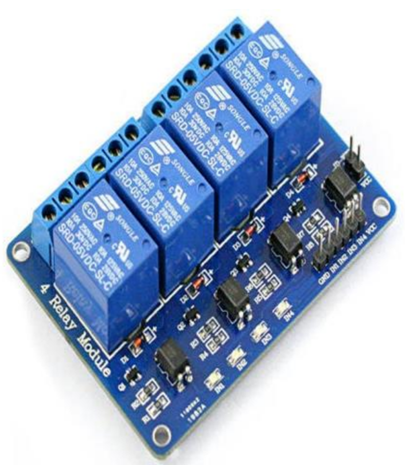
If either of the relays does not turn on:

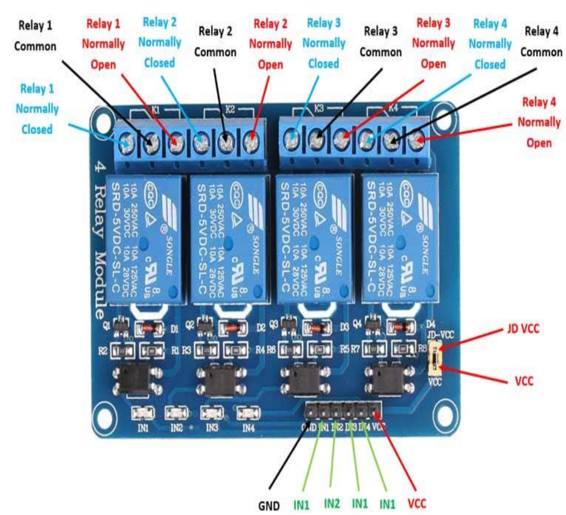
1. The contacts might be welded due to overcurrent/arcing. Shaking the module firmly might help unstick the contacts
2. The driver circuitry might have been damaged due to overvoltage.
3. Input polarity might be incorrect.
4. Jumper might not have been moved to the correct position

Dual-Channel Relay Module Applications

- Switching mains loads
- Home automation
- Battery backup
- High current load switching

5V Four-Channel Relay Module





5V Four-Channel Relay Module **5v Four-Channel Relay Module**

The **four-channel relay module** contains four 5V relays and the associated switching and isolating components, which makes interfacing with a microcontroller or sensor easy with minimum components and connections. The contacts on each relay are specified for 250VAC and 30VDC and 10A in each case, as marked on the body of the relays.

Four-Channel Relay Module Pinout

Pin Number	Pin Name	Description
1	GND	Ground reference for the module
2	IN1	Input to activate relay 1
3	IN2	Input to activate relay 2
4	IN3	Input to activate relay 3
5	IN4	Input to activate relay 4
6	V_{cc}	Power supply for the relay module
7	V_{cc}	Power supply selection jumper
8	JD-V_{cc}	Alternate power pin for the relay module

Components Present on A Four-Channel Relay Module

Following are the major components present on the four-channel relay module, we will get into the details of this later in the article.

5V relay, terminal blocks, male headers, transistors, optocouplers, diodes, and LEDs.

Four-Channel Relay Module Specifications

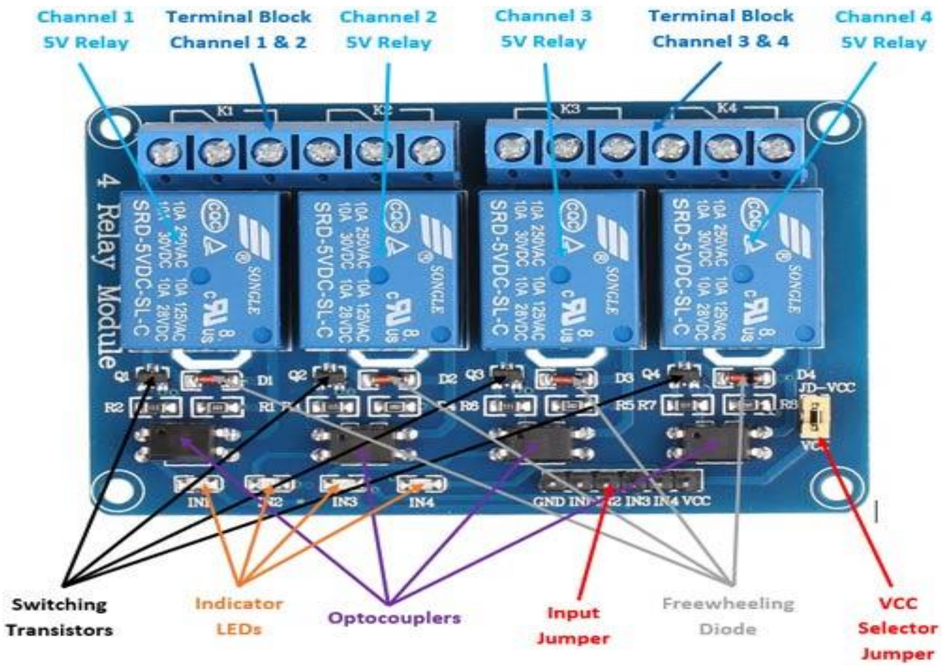
- Supply voltage – 3.75V to 6V
- Trigger current – 5mA
- Current when the relay is active - ~70mA (single), ~300mA (all four)
- Relay maximum contact voltage – 250VAC, 30VDC
- Relay maximum current – 10A

Alternate Relay Modules

Single-channel relay module, dual-channel relay module, eight-channel relay module.

Alternate Modules

SCRs, TRIACs, Solid State Relay module.



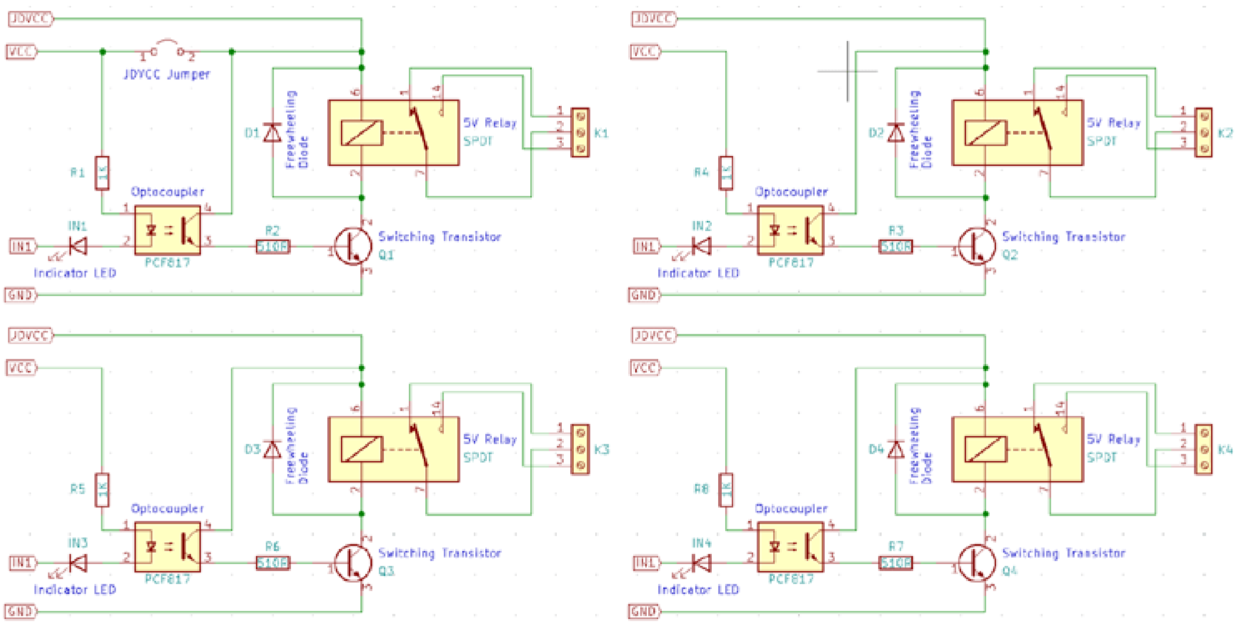
The four-channel relay module contains four 5V relays and the associated switching and isolating components, which makes interfacing with a microcontroller or sensor easy with minimum components and connections. There are two terminal blocks with six terminals each, and each block is shared by two relays. The terminals are screw type, which makes connections to mains wiring easy and changeable.

The four relays on the module are rated for 5V, which means the relay is activated when there is approximately 5V across the coil. The contacts on each relay are specified for 250VAC and 30VDC and 10A in each case, as marked on the body of the relays.

The switching transistors act as a buffer between the relay coils that require high currents, and the inputs which don't draw much current. They amplify the input signal so that they can drive the coils to activate the relays. The freewheeling diodes prevent voltage spikes across the transistors when the relay is turned off since the coils are an inductive load. The indicator LEDs glow when the coil of the respective relay is energized, indicating that the relay is active. The optocouplers form an additional layer of isolation between the load being switched and the inputs. The isolation is optional and can be selected using the V_{cc} selector jumper. The input jumper contains the main V_{cc}, GND, and input pins for easy connection using female jumper wires.

Internal Circuit Diagram For Four-Channel Relay Module

The circuit on the board is as follows:



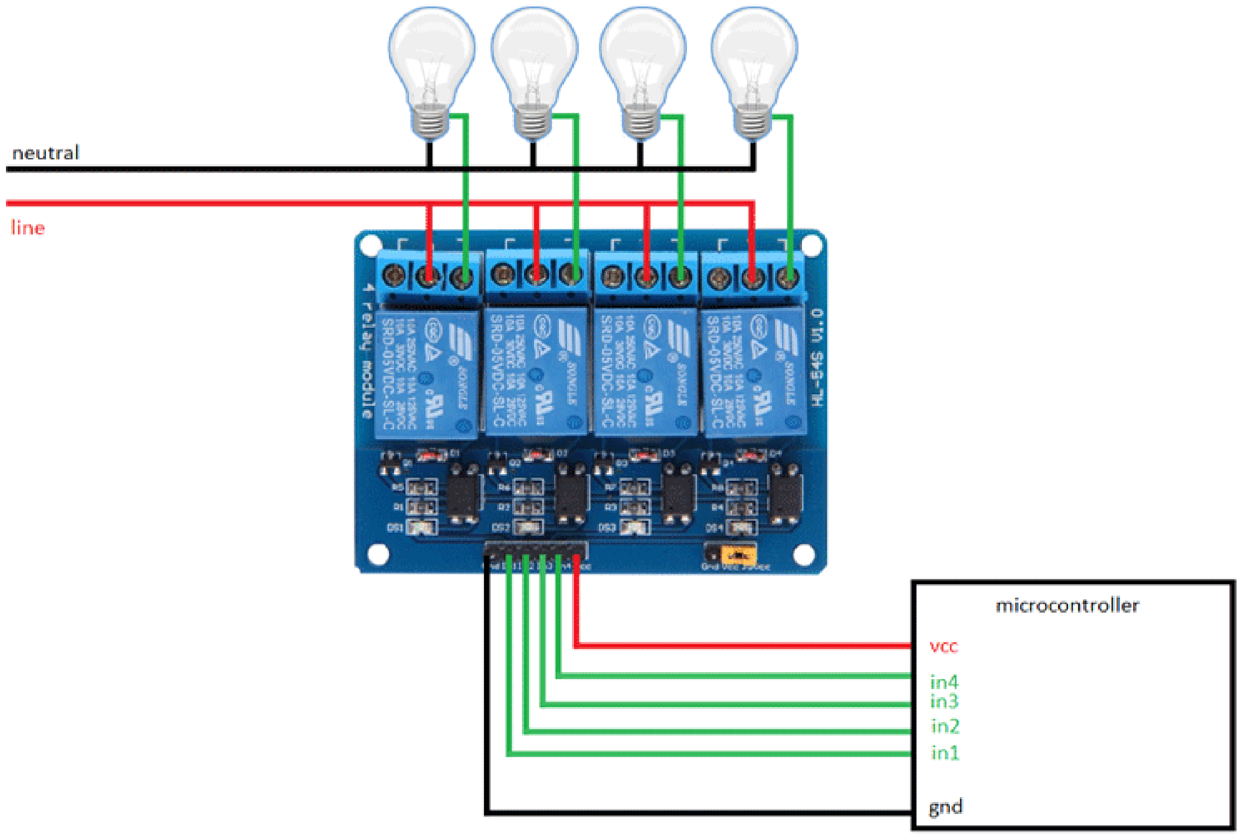
Each relay on the board has the same circuit, and the input ground is common to all four channels.

The driver circuit for this relay module is slightly different compared to traditional relay driving circuits since there is an optional additional layer of isolation. When the jumper is shorted, the relay and the input share the same V_{cc}, and when it is open, a separate power supply must be provided to the JD-V_{cc} jumper to power the relay coil and optocoupler output.

The inputs for this module are active low, meaning that the relay is activated when the signal on the input header is low. This is because the indicator LED and the input of the optocoupler are connected in series to the V_{cc} pin on one end, so the other end must be connected to the ground to enable the current flow. The optocouplers used here are the PCF817, which is a common optocoupler and can also be found in through-hole packaging.

How To Use The Four-Channel Relay Module

The four-channel can be used to switch multiple loads at the same time since there are four relays on the same module. This is useful in creating a central hub from where multiple remote loads can be powered. It is useful for tasks like home automation where the module can be placed in the main switchboard and can be connected to loads in other parts of the house and can be controlled from a central location using a microcontroller.



In this diagram, four separate loads (represented by lightbulbs) have been connected to the NO terminals of the relay. The live wire has been connected to the common terminal of each relay. When the relays are activated, the load is connected to the live wire and is powered. This setup can be reversed by connecting the load to the NC terminal that keeps it powered on till the relay is activated.

Dual-Channel Relay Module Basic Troubleshooting

If either of the relays does not turn on:

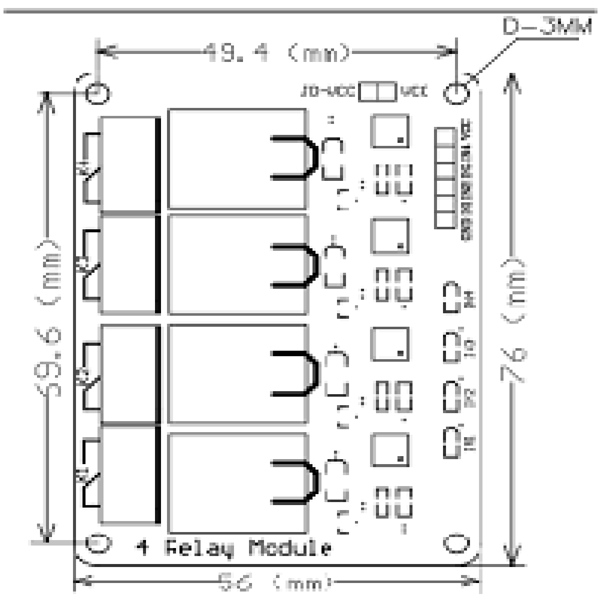
1. The contacts might be welded due to overcurrent/arcing. Shaking the module firmly might help unstick the contacts
2. The driver circuitry might have been damaged due to overvoltage.
3. Input polarity might be incorrect.
4. Jumper might not have been moved to the correct position

Dual-Channel Relay Module Applications

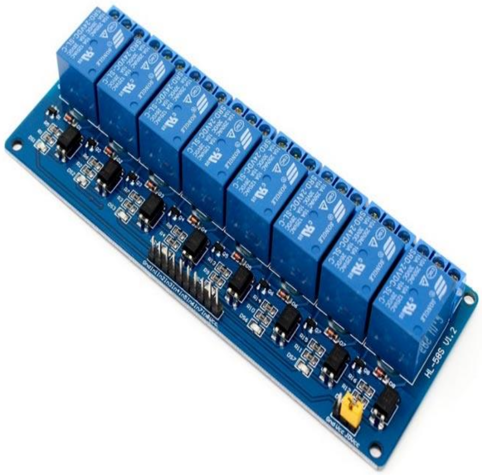
- Switching mains loads
- Home automation
- Battery backup
- High current load switching

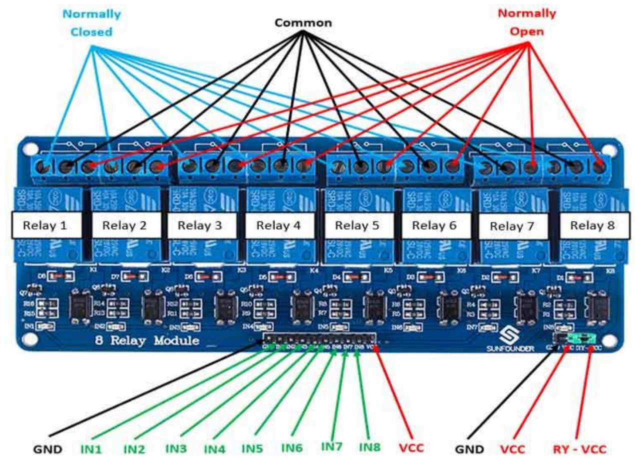
2D Model Of The Module

The **dimensions of the 4-Channel Relay module** are given below.



5V 8-Channel Relay Module





5V Eight-Channel Relay Module
Eight-Channel Relay Module

The eight-channel relay module contains eight 5V relays and the associated switching and isolating components, which makes interfacing with a microcontroller or sensor easy with minimum components and connections. Each relay on the board has the same circuit, and the input ground is common to all eight channels.

Eight-Channel Relay Module Pinout

Pin Number	Pin Name	Description
1	GND	Ground reference for the module
2	IN1	Input to activate relay 1
3	IN2	Input to activate relay 2
4	IN3	Input to activate relay 3
5	IN4	Input to activate relay 4
6	IN5	Input to activate relay 5
7	IN6	Input to activate relay 6

8	IN7	Input to activate relay 7
9	IN8	Input to activate relay 8
10	V_{cc}	Power supply for the relay module
11	GND	Ground reference for the module
12	V_{cc}	Power supply selection jumper
13	RY-V_{cc}	Alternate power pin for the relay module

Components Present on an Eight-Channel Relay Module

The following are the major components present on the eight-channel relay module, which will be described in detail later in the article.

5V relay, terminal blocks, male headers, transistors, optocouplers, diodes, and LEDs.

Eight-Channel Relay Module Specifications

- Supply voltage – 3.75V to 6V
- Trigger current – 5mA
- Current when relay is active - ~70mA (single), ~600mA (all eight)
- Relay maximum contact voltage – 250VAC, 30VDC
- Relay maximum current – 10A

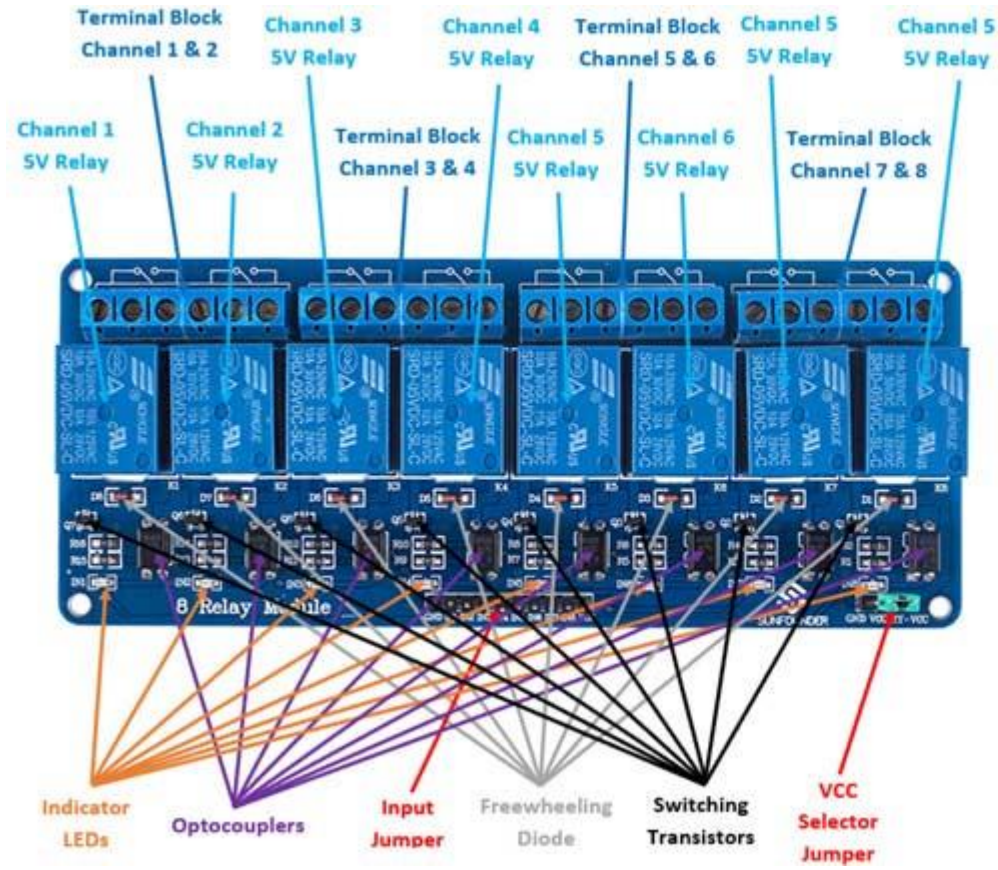
Alternate Relay Modules

Single-channel relay module, dual-channel relay module, four-channel relay module.

Alternate Modules

SCRs, TRIACs, Solid State Relay module.

Understanding 5V Eight-Channel Relay Module



The eight-channel relay module contains eight 5V relays and the associated switching and isolating components, which makes interfacing with a microcontroller or sensor easy with minimum components and connections. There are eight terminal blocks with six terminals each, and each block is shared by two relays. The terminals are screw type, which makes connections to mains wiring easy and changeable.

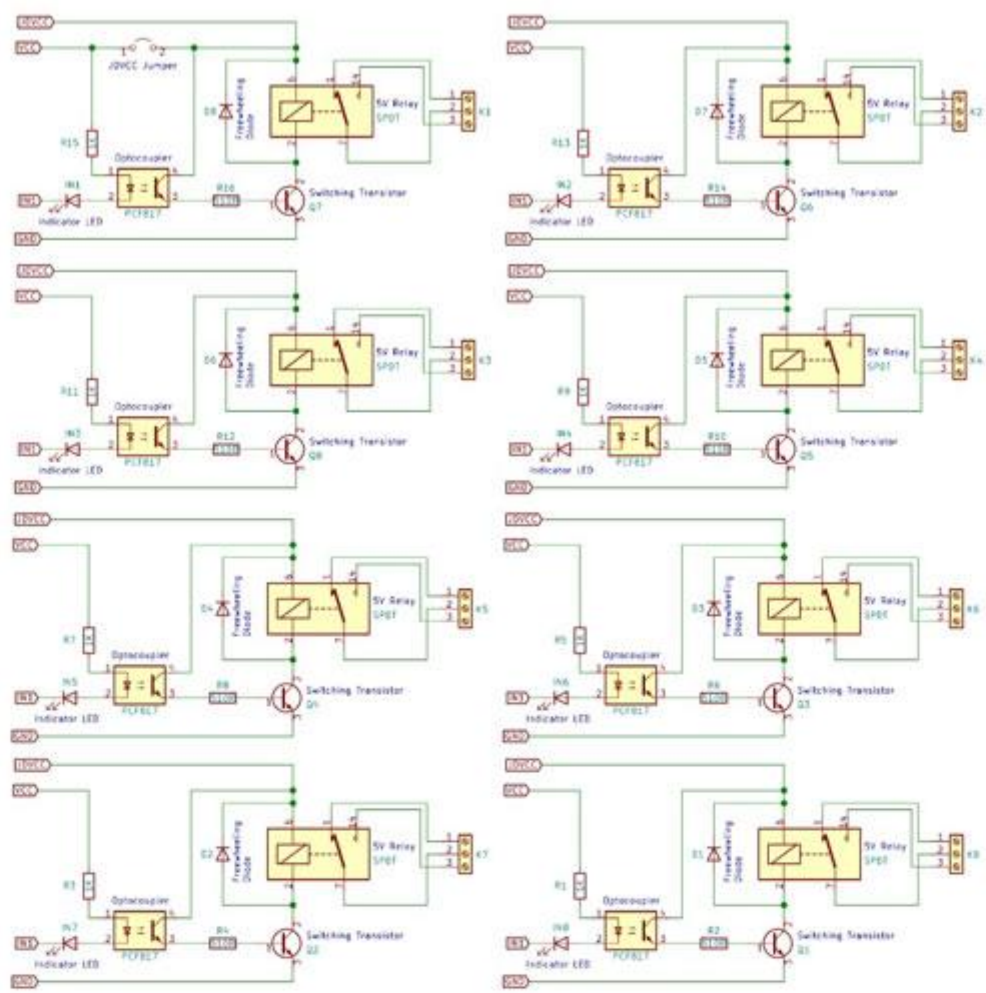
The eight relays on the module are rated for 5V, which means the relay is activated when there is approximately 5V across the coil. The contacts on each relay are specified for 250VAC and 30VDC and 10A in each case, as marked on the body of the relays. The switching transistors act as a buffer between the relay coils that require high currents, and the inputs which don’t draw much current. They amplify the input signal so that they can drive the coils to activate the relays.

The freewheeling diodes prevent voltage spikes across the transistors when the relay is turned off since the coils are an inductive load. The indicator LEDs glow when the coil of the respective relay is energized, indicating that the relay is active.

The optocouplers form an additional layer of isolation between the load being switched and the inputs. The isolation is optional and can be selected using the VCC selector jumper. The input jumper contains the main VCC, GND, and input pins for easy connection using female jumper wires.

Internal Circuit Diagram for Eight-Channel Relay Module

The circuit on the board is as follows:



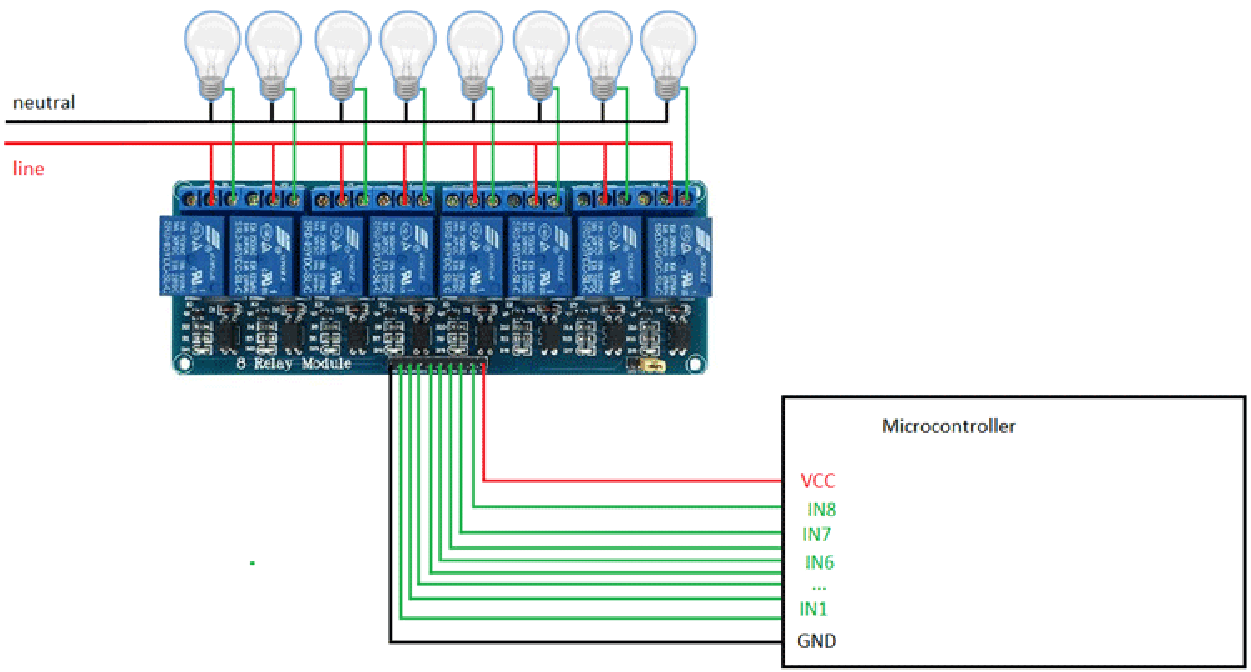
Each relay on the board has the same circuit, and the input ground is common to all eight channels.

The driver circuit for this relay module is slightly different compared to traditional relay driving circuits since there is an optional additional layer of isolation. When the jumper is shorted, the relay and the input share the same VCC, and when it is open, a separated power supply must be provided to the JD-VCC jumper to power the relay coil and optocoupler output.

The inputs for this module are active low, meaning that the relay is activated when the signal on the input header is low. This is because the indicator LED and the input of the optocoupler is connected in series to the VCC pin on one end, so the other end must be connected to the ground to enable the current flow. The optocoupler used here is the PCF817, which is a common optocoupler that can also be found in through-hole packaging.

How to Use the Eight-Channel Relay Module

The eight-channel can be used to switch multiple loads at the same time since there are eight relays on the same module. This is useful in creating a central hub from where multiple remote loads can be powered, which is useful for tasks like home automation where the module can be placed in the main switchboard and can be connected to loads in other parts of the house and can be controlled from a central location using a microcontroller.



In this diagram, eight separate loads (represented by lightbulbs) have been connected to the NO terminals of the relay. The live wire has been connected to the common terminal of each relay. When the relays are activated, the load is connected to the live wire and is powered. This setup can be reversed by connecting the load to the NC terminal, which will keep it powered on till the relay is activated.

Eight-Channel Relay Module Basic Troubleshooting

If either of the relays does not turn on:

1. The contacts might be welded due to overcurrent/arcing. Shaking the module firmly might help unstick the contacts.
2. The driver circuitry might have been damaged due to overvoltage.

3. Input polarity might be incorrect.
4. Jumper might not have been moved to the correct position

Eight-Channel Relay Module Applications

- Switching mains loads
- Home automation
- Battery backup
- High current load switching